Physical Chemistry (I)

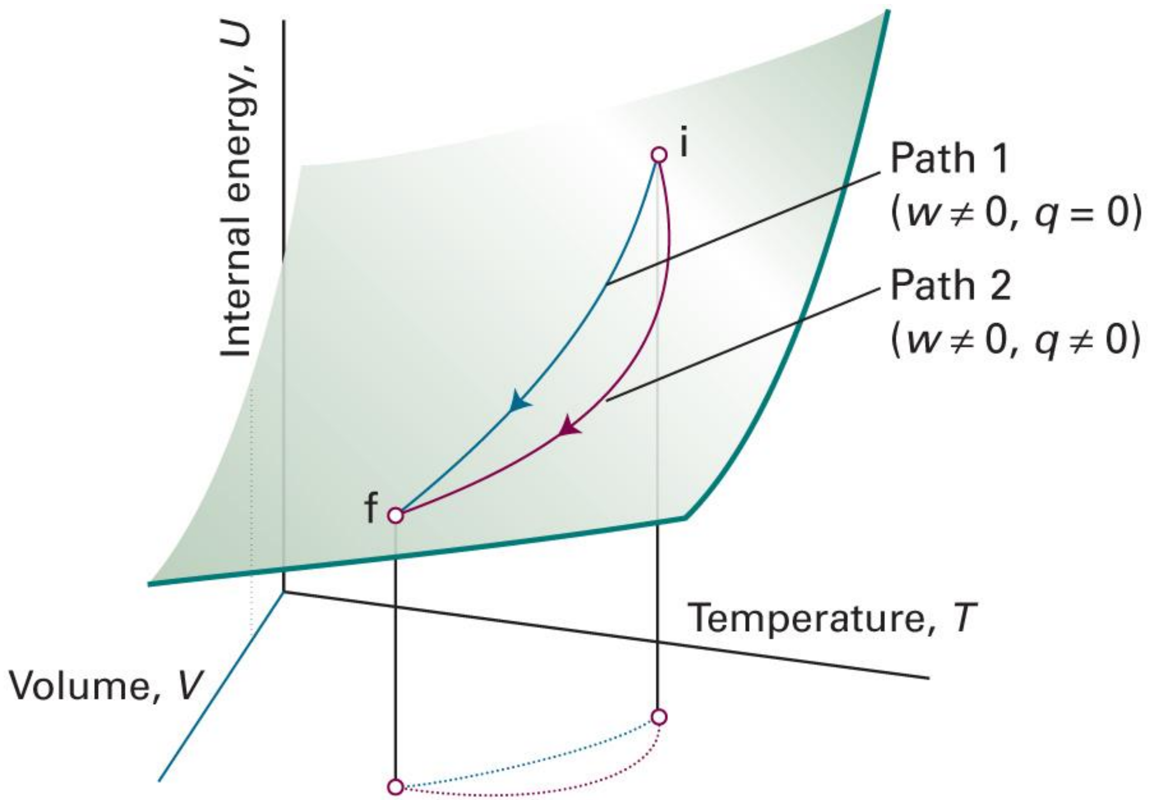
Lecture 07

In the Last Class

- Differential form of internal energy
  - State and path functions
  - Applications
- Enthalpy
  - Definition and physical meaning
  - Differential form (cf. response functions)
  - C_V and C_p
- Adiabatic changes

Review

State and Path Functions



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2D.1)

Differential Form of U

For $U = U(V,T)$,

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

$$dU = \pi_T dV + C_V dT$$

- Can calculate other variables such as $(\partial U/\partial T)_p$

Response Functions

- Expansion coefficient

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$$

- Isothermal compressibility

$$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$$

Enthalpy $\boldsymbol{H}$

$$H = U + pV$$

- Energy required to create a system (U) and to make a room for the system (pV)

- Meaning: $dH = \delta q_p$ or $\Delta H = q_p$

$$C_p = \left(\frac{\partial H}{\partial T}\right)_p$$

Differential Form of $\boldsymbol{H}$

For $H = H(p, T)$,

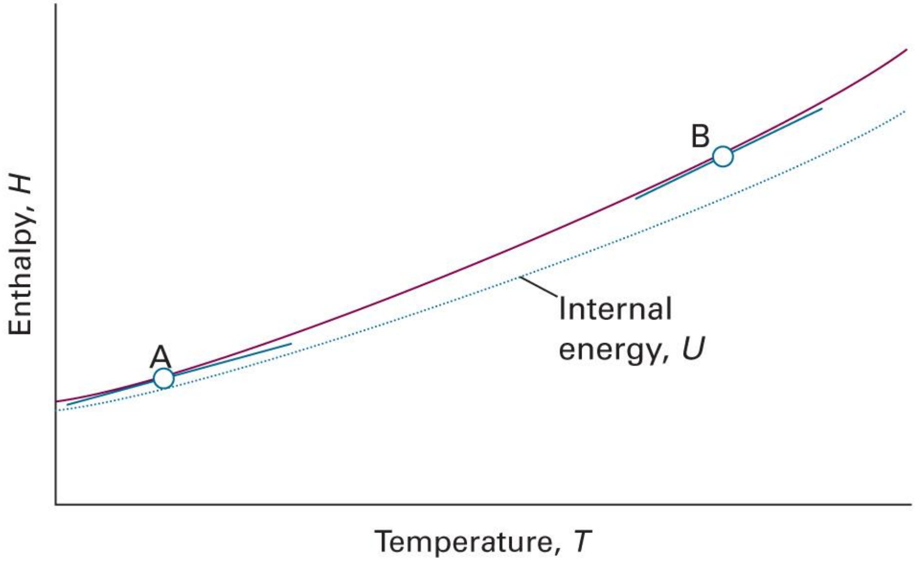
$$dH = \left(\frac{\partial H}{\partial p}\right)_T dp + \left(\frac{\partial H}{\partial T}\right)_p dT$$

$$dH = -C_p \mu dp + C_p dT$$

(Compare: $dU = \pi_T dV + C_V dT$)

C_V and C_p

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V, \qquad C_p = \left(\frac{\partial H}{\partial T}\right)_p$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2B.3)

Adiabatic Changes

$$\delta q = 0$$

For a perfect gas whose C_V is constant,

- Work: $w = C_V\left(T_f - T_i\right)$
- Relations in a reversible change:
  - $VT^c = \text{constant}\ (c = C_V/nR)$
  - $pV^\gamma = \text{constant}\ (\gamma = 1 + 1/c)$

Thermochemistry

"The study of the energy transferred as heat during the course of chemical reactions"

$$\underbrace{\mathrm{A + B}}_{\text{state } i} \longrightarrow \underbrace{\mathrm{C + D}}_{\text{state } f}$$

$$q_V = \Delta U$$

$$q_p = \Delta H$$

Endothermic and Exothermic

$$\mathrm{A} + \mathrm{B} \longrightarrow \mathrm{C} + \mathrm{D},\ \Delta H = q_p$$

$\Delta H > 0$: endothermic reaction

$\Delta H < 0$: exothermic reaction

Standard Enthalpy, $\boldsymbol{H}^{\ominus}$

Enthalpy H of a substance in its standard state

- Standard state of a substance: its pure form at 1 bar
  - Defined for a specific temperature
  - If temperature is not specified, use the conventional temperature of 298.15 K
- Standard enthalpy change $\Delta H^{\ominus} = H_f^{\ominus} - H_i^{\ominus}$

Example: Vaporization of Water

$$H_2O(l) \rightarrow H_2O(g)$$

$$\Delta H^{\ominus}(373\ \mathrm{K}) = 40.66\ \mathrm{kJ\ mol^{-1}}$$

pure liquid water → pure gas water

at 1 bar and 373 K

Not in Atkins

$\boldsymbol{H}$ and $\Delta\boldsymbol{H}$

$$\Delta H = H(\text{product}) - H(\text{reactant})$$

$$H = U + pV$$

One can determine absolute H for a perfect gas

But what about <u>real</u> substances?

(liquids, solids, solutions, ...)

Instead, we focus on the "change," ΔH, in chemistry

For Typical Changes

- The standard enthalpy change is called the standard enthalpy of [transition]
- Physical change
  - Vaporization $\Delta_{\mathrm{vap}}H^{\ominus}$: standard enthalpy of vaporization
  - Fusion $\Delta_{\mathrm{fus}}H^{\ominus}$: standard enthalpy of fusion
- Chemical change
  - Reaction $\Delta_{\mathrm{r}}H^{\ominus}$: standard enthalpy of reaction
  - Combustion $\Delta_{\mathrm{c}}H^{\ominus}$: standard enthalpy of combustion

Several Types of Enthalpies

Transition	Process	Symbol*
Transition	Phase $\alpha \rightarrow$ phase β	$\Delta_{trs}H$
Fusion	$s \rightarrow l$	$\Delta_{fus}H$
Vaporization	$l \rightarrow g$	$\Delta_{vap}H$
Sublimation	$s \rightarrow g$	$\Delta_{sub}H$
Mixing	Pure $\rightarrow$ mixture	$\Delta_{mix}H$
Solution	Solute $\rightarrow$ solution	$\Delta_{sol}H$
Hydration	$X^{\pm}(g) \rightarrow X^{\pm}(aq)$	$\Delta_{hyd}H$
Atomization	Species(s, l, g) $\rightarrow$ atoms(g)	$\Delta_{at}H$
Ionization	$X(g) \rightarrow X^{+}(g) + e^{-}(g)$	$\Delta_{ion}H$
Electron gain	$X(g) + e^{-}(g) \rightarrow X^{-}(g)$	$\Delta_{eg}H$
Reaction	Reactants $\rightarrow$ products	$\Delta_{r}H$
Combustion	Compound(s, l, g) + $O_2(g) \rightarrow CO_2(g) + H_2O(l, g)$	$\Delta_{c}H$
Formation	Elements $\rightarrow$ compound	$\Delta_{f}H$
Activation	Reactants $\rightarrow$ activated complex	$\Delta^{\ddagger}H$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Table 2C.2)

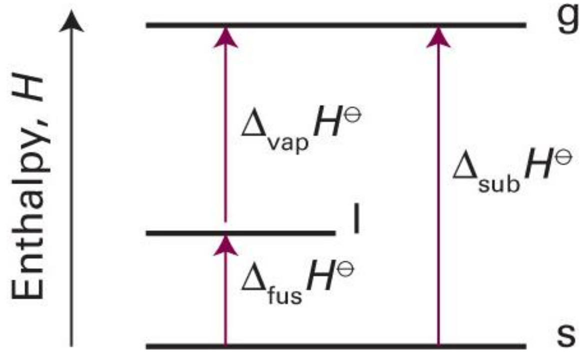
H as State Function

ΔH is always same if the initial and final states are same

Path 1: $H_2O(s) \longrightarrow H_2O(g)$, $\Delta_{\mathrm{sub}}H^{\ominus}$

Path 2: (1) $H_2O(s) \longrightarrow H_2O(l)$, $\Delta_{\mathrm{fus}}H^{\ominus}$

(2) $H_2O(l) \longrightarrow H_2O(g)$, $\Delta_{\mathrm{vap}}H^{\ominus}$

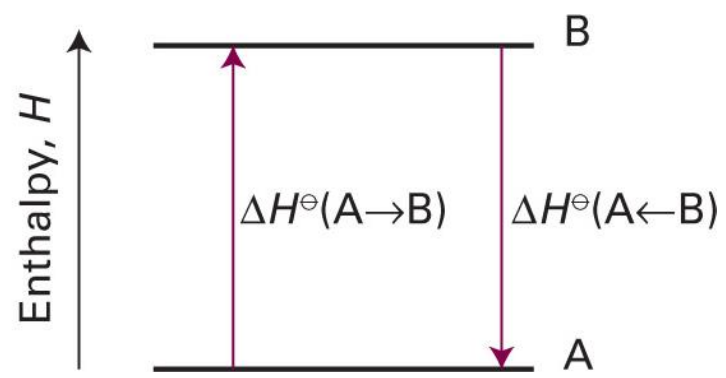


$$\Delta_{\mathrm{sub}}H^{\ominus} = \Delta_{\mathrm{fus}}H^{\ominus} + \Delta_{\mathrm{vap}}H^{\ominus}$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., p. 50)

H as State Function

$$\Delta H^{\ominus}(\text{forward}) = -\Delta H^{\ominus}(\text{reverse})$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., p. 50)

Stoichiometry: Mole Method

$$\nu_{\mathrm{A}}\mathrm{A} + \nu_{\mathrm{B}}\mathrm{B} \longrightarrow \nu_{\mathrm{C}}\mathrm{C} + \nu_{\mathrm{D}}\mathrm{D}$$

means

"ν_{A} moles of A and ν_{B} moles of B react

to produce ν_{C} moles of C and ν_{D} moles of D."

(ν_i: stoichiometric coefficients)

Standard Reaction Enthalpy

$$\nu_{\mathrm{A}}\mathrm{A} + \nu_{\mathrm{B}}\mathrm{B} \longrightarrow \nu_{\mathrm{C}}\mathrm{C} + \nu_{\mathrm{D}}\mathrm{D}$$

- Standard reaction enthalpy $\Delta_{\mathrm{r}}H^{\ominus}$: standard enthalpy change $\Delta H^{\ominus}$ for the reaction per …
  - ν_{A} moles of consumed A
  - ν_{B} moles of consumed B
  - ν_{C} moles of produced C
  - ν_{D} moles of produced D

Example

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

$$\Delta_r H^{\ominus} = -890 \text{ kJ mol}^{-1}$$

The enthalpy change is -890 kJ at 1 bar and 298.15 K per

- 1 mol of $CH_4(g)$ in its pure form
- 2 mol of $O_2(g)$ in its pure form
- 1 mol of $CO_2(g)$ in its pure form
- 2 mol of $H_2O(l)$ in its pure form

Formal Definition

$$\Delta_\mathrm{r}H^{\ominus} = \sum_{\text{products } i} \nu(i)H_m^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j)H_m^{\ominus}(j)$$

e.g. $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$

$$\Delta_\mathrm{r}H^{\ominus} = \nu(\mathrm{CO_2})H_m^{\ominus}(\mathrm{CO_2}, g) + \nu(\mathrm{H_2O})H_m^{\ominus}(\mathrm{H_2O}, l)$$

$$-\nu(\mathrm{CH_4})H_m^{\ominus}(\mathrm{CH_4}, g) - \nu(\mathrm{O_2})H_m^{\ominus}(\mathrm{O_2}, g)$$

$$= H_m^{\ominus}(\mathrm{CO_2}, g) + 2H_m^{\ominus}(\mathrm{H_2O}, l) - H_m^{\ominus}(\mathrm{CH_4}, g) - 2H_m^{\ominus}(\mathrm{O_2}, g)$$

Problem in $\Delta_{\mathrm{r}}\boldsymbol{H}^{\boldsymbol{\ominus}}$

$$\Delta_{\mathrm{r}}H^{\ominus} = \sum_{\text{products } i} \nu(i)H_m^{\ominus}(i) - \sum_{\text{reactants } j} \nu(j)H_m^{\ominus}(j)$$

Can we determine absolute $H_m^{\ominus}(i)$?

Hess's Law

"The standard reaction enthalpy is

the sum of the values for the individual reactions

into which the overall reaction may be divided."

This holds because H is a state function

Example 2C.1

The standard reaction enthalpy for the hydrogenation of propene, $CH_2{=}CHCH_3(g) + H_2(g) \rightarrow CH_3CH_2CH_3(g)$, is -124 kJ mol^{-1}.

The standard reaction enthalpy for the combustion of propane, $CH_3CH_2CH_3(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$, is -2220 kJ mol^{-1}.

The standard reaction enthalpy for the formation of water, $H_2(g) + (1/2)O_2(g) \rightarrow H_2O(l)$, is -286 kJ mol^{-1}.

Calculate the standard enthalpy of combustion of propene.

The target reaction is

$$CH_2{=}CHCH_3(g) + (9/2)O_2(g) \rightarrow 3CO_2(g) + 3H_2O(l)$$

Example 2C.1

$$CH_2{=}CHCH_3(g) + (9/2)O_2(g) \rightarrow 3CO_2(g) + 3H_2O(l)$$

Arranging the given reactions, we can obtain

$$CH_2{=}CHCH_3(g) + H_2(g) \rightarrow CH_3CH_2CH_3(g),\ \Delta_r H^{\ominus}(1) = -124 \text{ kJ mol}^{-1}$$

$$CH_3CH_2CH_3(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l),\ \Delta_r H^{\ominus}(2) = -2220 \text{ kJ mol}^{-1}$$

$$H_2O(l) \rightarrow H_2(g) + (1/2)O_2(g),\ \Delta_r H^{\ominus}(3) = +286 \text{ kJ mol}^{-1}$$

Summing up, the whole reaction is

$$CH_2{=}CHCH_3(g) + (9/2)O_2(g) \rightarrow 3CO_2(g) + 3H_2O(l)$$

and

$$\Delta_r H^{\ominus} = \Delta_r H^{\ominus}(1) + \Delta_r H^{\ominus}(2) + \Delta_r H^{\ominus}(3) = -2058 \text{ kJ mol}^{-1}$$

Standard Enthalpy of Formation

- Standard enthalpy of formation $\Delta_\mathrm{f}H^\ominus$ of a substance: $\Delta_\mathrm{r}H^\ominus$ for the formation of the compound from its elements in their reference states
- Reference state of an element: its most stable state at the specified temperature and 1 bar

e.g. $\Delta_\mathrm{f}H^\ominus$ of liquid benzene at 298 K: +49.0 kJ mol^{-1}

$$6C(s, \text{graphite}) + 3H_2(g) \longrightarrow C_6H_6(l)$$

$\Delta_f \boldsymbol{H}^{\ominus}$ of Elements

- For an element …
  - in its reference state, $\Delta_f H^{\ominus} = 0$ (for all T)

  C(s, graphite) ⟶ C(s, graphite), $\Delta_f H^{\ominus}$ = 0

  - in the other state, $\Delta_f H^{\ominus} > 0$

  C(s, graphite) ⟶ C(s, diamond), $\Delta_f H^{\ominus}$(298 K) = 1.895 kJ mol^{-1}

- Reference state of phosphorus (P): white phosphorus
  - not the most stable, but the most reproducible

  P(s, white) ⟶ P(s, red), $\Delta_f H^{\ominus}$(298 K) = -17.46 kJ mol^{-1}

$\Delta_f H^\ominus$ of Ions

- $\Delta_f H^\ominus$ of ions: *e.g.* $Na(s) \longrightarrow Na^+(aq)$?

Impossible to isolate $Na^+(aq)$

- Define $\Delta_f H^\ominus$ for one ion, and use it to determine others

$$\Delta_f H^\ominus(H^+, aq) = 0$$

e.g. $HBr(aq) = H^+(aq) + Br^-(aq)$

$$\underbrace{\Delta_f H^\ominus(HBr, aq)}_{\text{measurable}} = \underbrace{\Delta_f H^\ominus(H^+, aq)}_{=0} + \Delta_f H^\ominus(Br^-, aq)$$

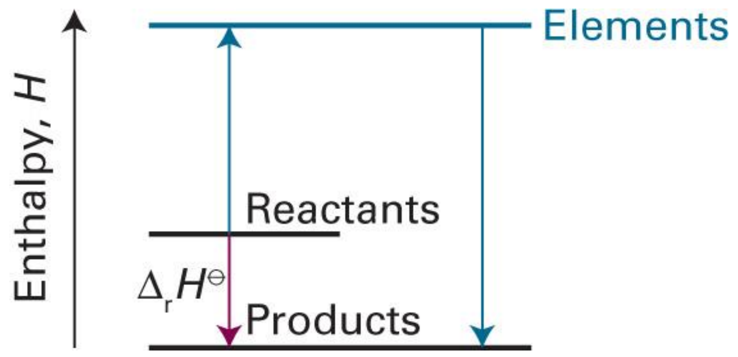
Some Values of $\Delta_f H^\ominus$

	$\Delta_f H^\ominus/(\text{kJ mol}^{-1})$
$H_2O(l)$	−285.83
$H_2O(g)$	−241.82
$NH_3(g)$	−46.11
$N_2H_4(l)$	+50.63
$NO_2(g)$	+33.18
$N_2O_4(g)$	+9.16
$NaCl(s)$	−411.15
$KCl(s)$	−436.75

	$\Delta_f H^\ominus/(\text{kJ mol}^{-1})$
$CH_4(g)$	−74.81
$C_6H_6(l)$	+49.0
$C_6H_{12}(l)$	−156
$CH_3OH(l)$	−238.66
$CH_3CH_2OH(l)$	−277.69

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Tables 2C.4 and 2C.5)

$\Delta_f H^\ominus$ and $\Delta_r H^\ominus$



$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i)\Delta_f H^\ominus(i) - \sum_{\text{reactants } j} \nu(j)\Delta_f H^\ominus(j)$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., p. 55)

Temperature Dependence

$$dH = C_p dT$$

$$\int_{\text{state } i}^{\text{state } f} dH = \int_{\text{state } i}^{\text{state } f} C_p \, dT$$

$$H_f - H_i = \int_{T_i}^{T_f} C_p \, dT$$

$$H(T_f) - H(T_i) = \int_{T_i}^{T_f} C_p \, dT$$

Temperature Dependence

For ν_{A}A + ν_{B}B ⟶ ν_{C}C + ν_{D}D,

$$\Delta_{\mathrm{r}}H^{\ominus}(T_f) = \nu_{\mathrm{C}}H_m^{\ominus}(\mathrm{C},T_f) + \nu_{\mathrm{D}}H_m^{\ominus}(\mathrm{D},T_f) - \nu_{\mathrm{A}}H_m^{\ominus}(\mathrm{A},T_f) - \nu_{\mathrm{B}}H_m^{\ominus}(\mathrm{B},T_f)$$

Since $H(T_f) = H(T_i) + \int_{T_i}^{T_f} C_p\,dT$,

$$H_m^{\ominus}(T_f) = H_m^{\ominus}(T_i) + \int_{T_i}^{T_f} C_{p,m}^{\ominus}\,dT$$

$$\Delta_{\mathrm{r}}H^{\ominus}(T_f)$$

$$= \nu_{\mathrm{C}}\left\{H_m^{\ominus}(\mathrm{C},T_i) + \int_{T_i}^{T_f} C_{p,m}^{\ominus}(\mathrm{C})\,dT\right\} + \nu_{\mathrm{D}}\left\{H_m^{\ominus}(\mathrm{D},T_i) + \int_{T_i}^{T_f} C_{p,m}^{\ominus}(\mathrm{D})\,dT\right\}$$

$$- \nu_{\mathrm{A}}\left\{H_m^{\ominus}(\mathrm{A},T_i) + \int_{T_i}^{T_f} C_{p,m}^{\ominus}(\mathrm{A})\,dT\right\} - \nu_{\mathrm{B}}\left\{H_m^{\ominus}(\mathrm{B},T_i) + \int_{T_i}^{T_f} C_{p,m}^{\ominus}(\mathrm{B})\,dT\right\}$$

Temperature Dependence

$$\Delta_r H^{\ominus}(T_f) = \nu_C H_m^{\ominus}(C, T_i) + \nu_D H_m^{\ominus}(D, T_i) - \nu_A H_m^{\ominus}(A, T_i) - \nu_B H_m^{\ominus}(B, T_i)$$

$$+ \int_{T_i}^{T_f} \underbrace{\left\{\nu_C C_{p,m}^{\ominus}(\text{C}) + \nu_D C_{p,m}^{\ominus}(\text{D}) - \nu_A C_{p,m}^{\ominus}(\text{A}) - \nu_B C_{p,m}^{\ominus}(\text{B})\right\}}_{\Delta_r C_p^{\ominus}} dT$$

$$\Delta_r H^{\ominus}(T_f) = \Delta_r H^{\ominus}(T_f) + \int_{T_i}^{T_f} \Delta_r C_p^{\ominus}\, dT$$

Kirchhoff's Law

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \int_{T_i}^{T_f} \Delta_r C_p^\ominus \, dT$$

where

$$\Delta_r C_p^\ominus = \sum_{\text{products } i} \nu(i) C_{p,m}^\ominus(i) - \sum_{\text{reactants } j} \nu(j) C_{p,m}^\ominus(j)$$

If $\Delta_r C_p^\ominus$ is constant in $[T_i, T_f]$,

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \Delta_r C_p^\ominus \times (T_f - T_i)$$

Example 2C.2

The standard enthalpy of formation of $H_2O(g)$ at 298 K is -241.82 kJ mol^{-1}. Estimate its value at 100 °C given the following values of the molar heat capacities at constant pressure: $H_2O(g)$: 33.58 J K^{-1} mol^{-1}; $H_2(g)$: 28.84 J K^{-1} mol^{-1}; $O_2(g)$: 29.37 J K^{-1} mol^{-1}. Assume that the heat capacities are independent of temperature.

The formation reaction is $H_2(g) + (1/2)O_2(g) \rightarrow H_2O(g)$. Hence,

$$\Delta_r C_p^\ominus = C_{p,m}^\ominus(\mathrm{H_2O}, g) - C_{p,m}^\ominus(\mathrm{H_2}, g) - (1/2)C_{p,m}^\ominus(\mathrm{O_2}, g)$$

$$= 33.58\ \mathrm{J\ K^{-1}\ mol^{-1}} - 28.84\ \mathrm{J\ K^{-1}\ mol^{-1}} - (1/2)\times(29.37\ \mathrm{J\ K^{-1}\ mol^{-1}})$$

$$= -9.94\ \mathrm{J\ K^{-1}\ mol^{-1}}$$

Example 2C.2

Using Kirchhoff's law,

$$\Delta_r H^{\ominus}(T_f) = \Delta_r H^{\ominus}(T_i) + \Delta_r C_p^{\ominus}(T_f - T_i)$$

$$\Delta_r H^{\ominus}(373\ \mathrm{K}) = \Delta_r H^{\ominus}(298\ \mathrm{K}) + \Delta_r C_p^{\ominus} \times (373\ \mathrm{K} - 298\ \mathrm{K})$$

$= -241.82\ \mathrm{kJ\ mol^{-1}} + (-9.94\ \mathrm{J\ K^{-1}\ mol^{-1}}) \times (75\ \mathrm{K})$

$= -242.57\ \mathrm{kJ\ mol^{-1}}$

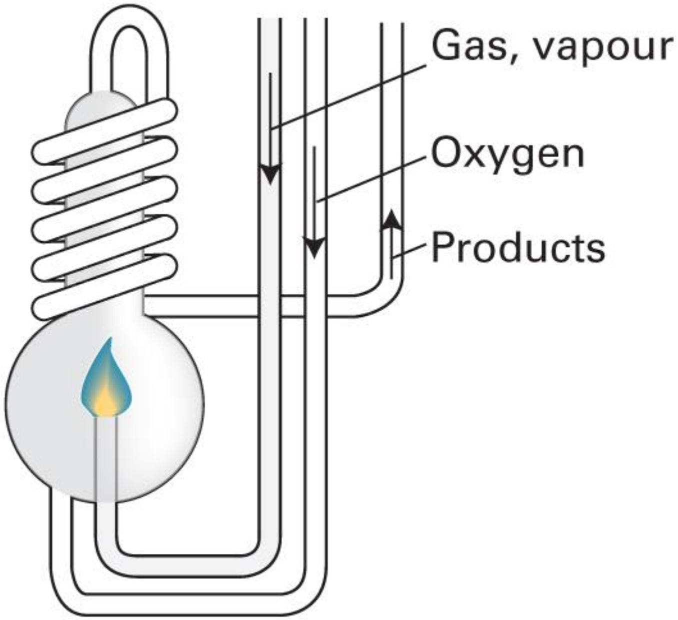
Measurement of Enthalpy

1. Adiabatic flame calorimeter

2. Differential scanning calorimeter (DSC)

3. Isothermal titration calorimeter (ITC)

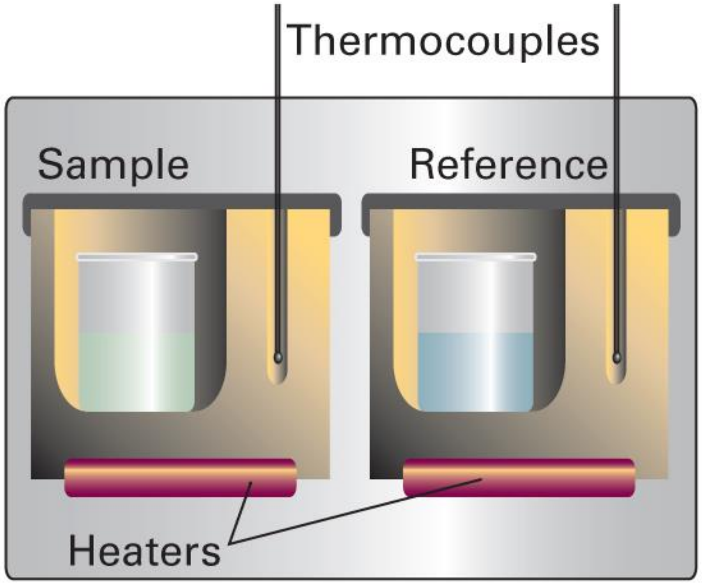
Adiabatic Flame Calorimeter

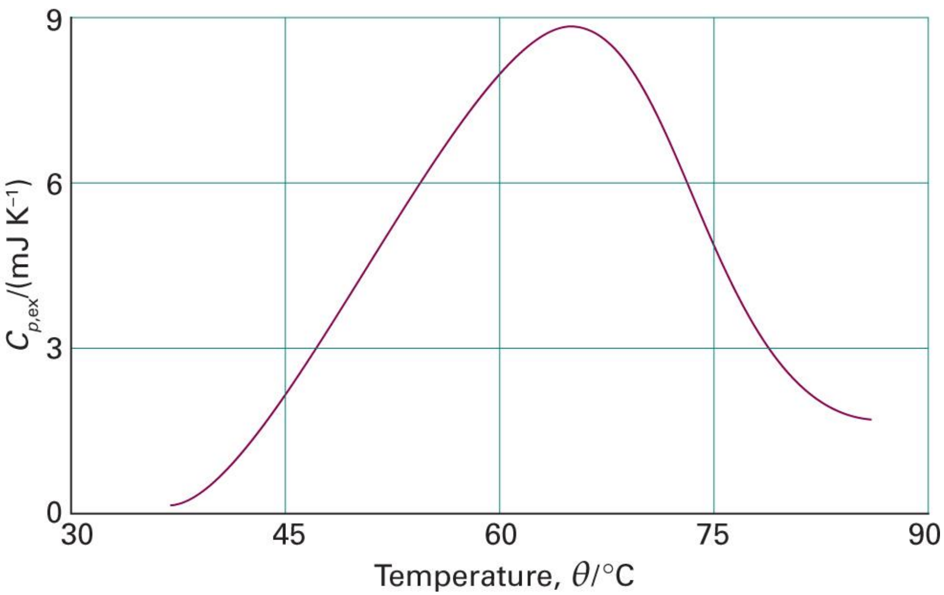


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2B.2)

Topic 2C

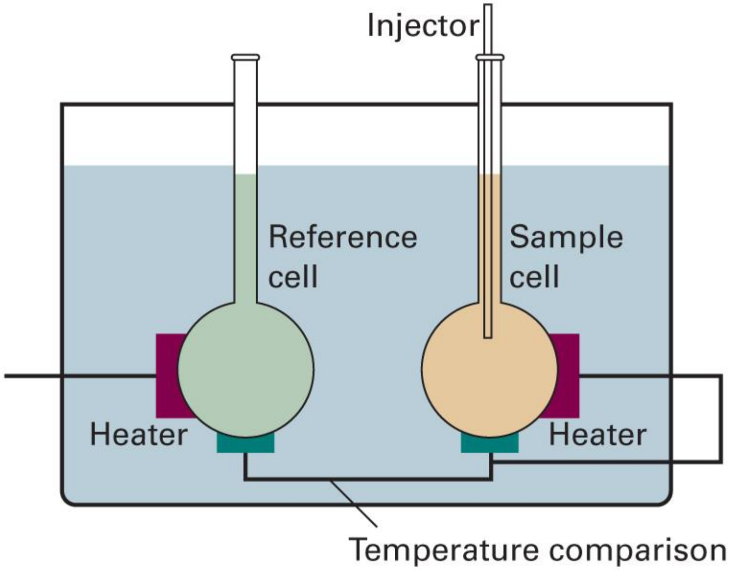
DSC

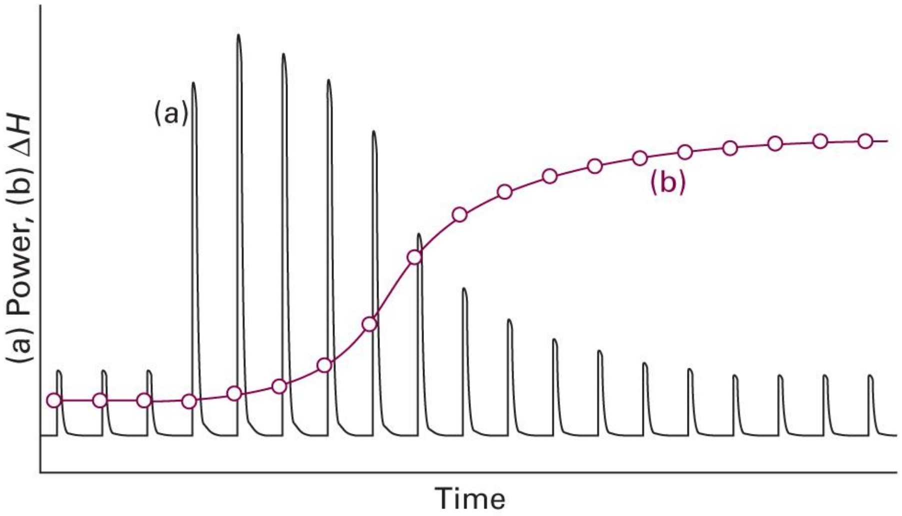




(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 2C.2 and 2C.3)

ITC





(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figures 2C.4 and 2C.5)

Summary for Chapter 3 (3)

Enthalpy

$H = U + pV$

Standard enthalpy $H^{\ominus}$

Standard enthalpy change $\Delta H^{\ominus}$

Heat capacity

$C_p = (\partial H / \partial T)_p$

Hess's law

Kirchhoff's law

Standard reaction enthalpy $\Delta_\mathrm{r} H^{\ominus}$

$$\Delta_\mathrm{r} H^{\ominus} = \sum_{\mathrm{product}\ i} \nu(i) H_m^{\ominus}(i) - \sum_{\mathrm{reactant}\ j} \nu(j) H_m^{\ominus}(j)$$

Standard enthalpy of formation $\Delta_\mathrm{f} H^{\ominus}$

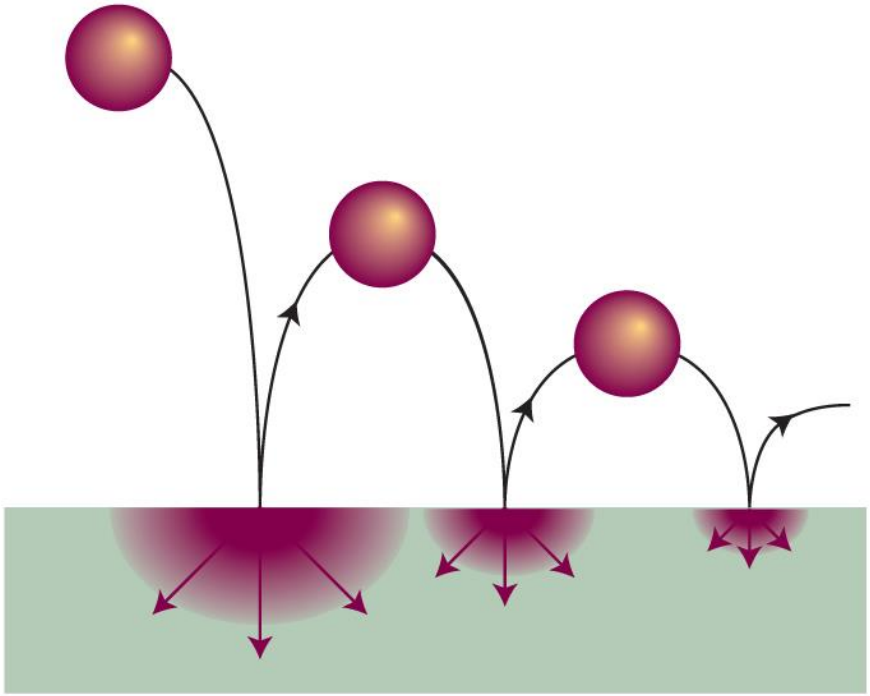
$$\Delta_\mathrm{r} H^{\ominus} = \sum_{\mathrm{product}\ i} \nu(i) \Delta_\mathrm{f} H^{\ominus}(i) - \sum_{\mathrm{reactant}\ j} \nu(j) \Delta_\mathrm{f} H^{\ominus}(j)$$

4. Entropy

Chapter Overview

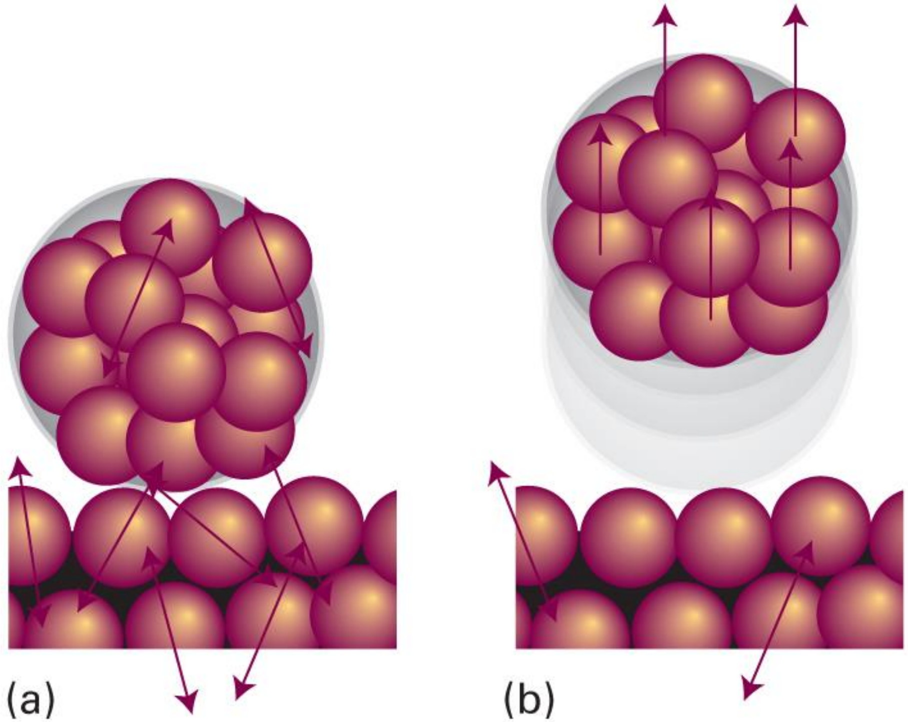
- Second law of thermodynamics
  - Kelvin's statement and Clausius's statement
  - Heat engine and Carnot cycle
- Entropy
  - Definition and properties
  - Third law of thermodynamics
  - Entropy and thermochemistry
- Physical meaning of entropy
  - Thermodynamic meaning
  - Statistical meaning: Boltzmann's formula

"Arrow of Time"



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.1)

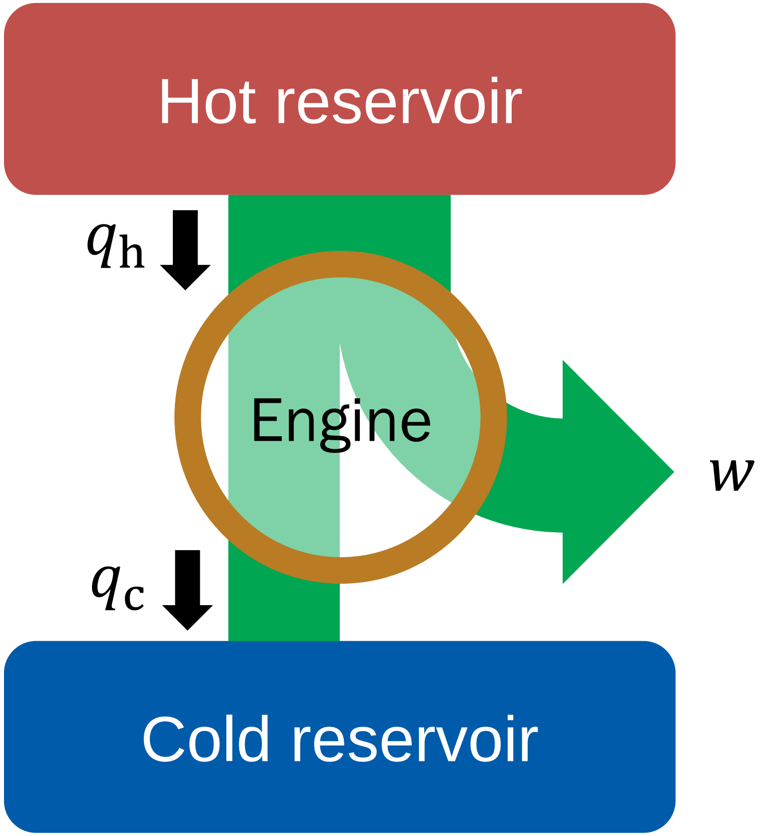
Why?



Heat and work are not equal

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 3A.2)

Heat Engine

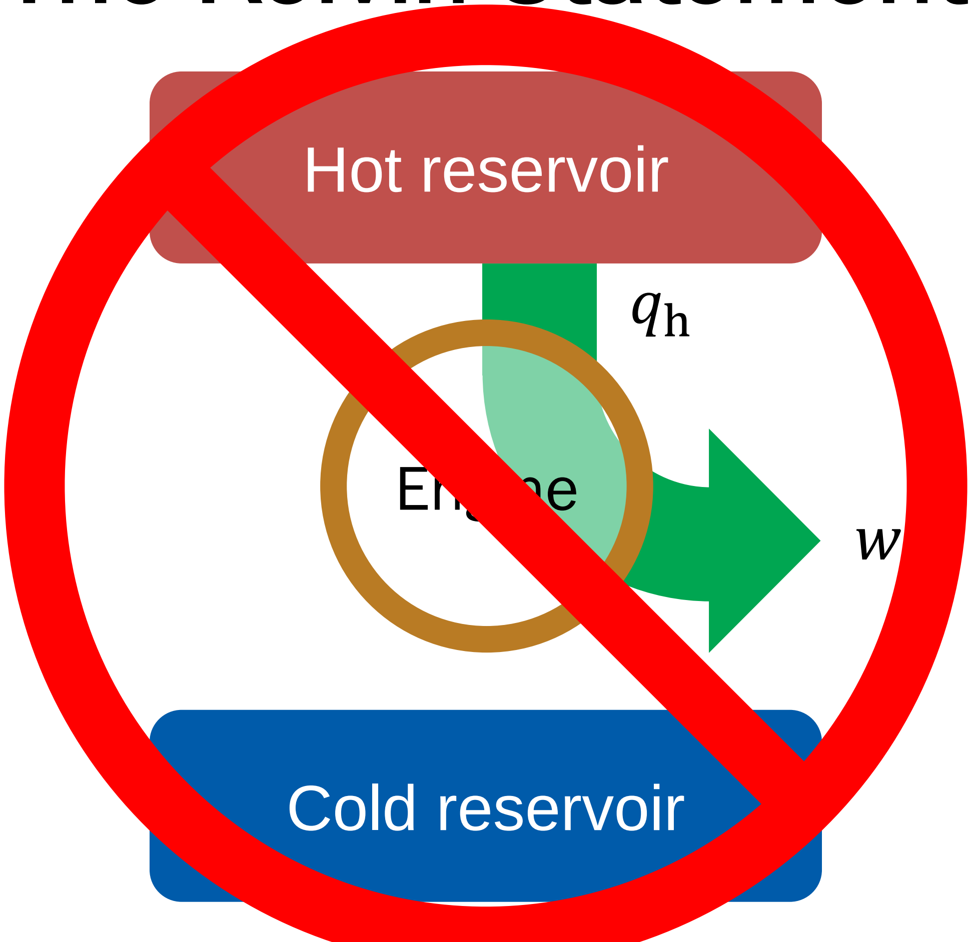


$$q_{\mathrm{h}} + q_{\mathrm{c}} + w = 0$$

Second Law (Kelvin)

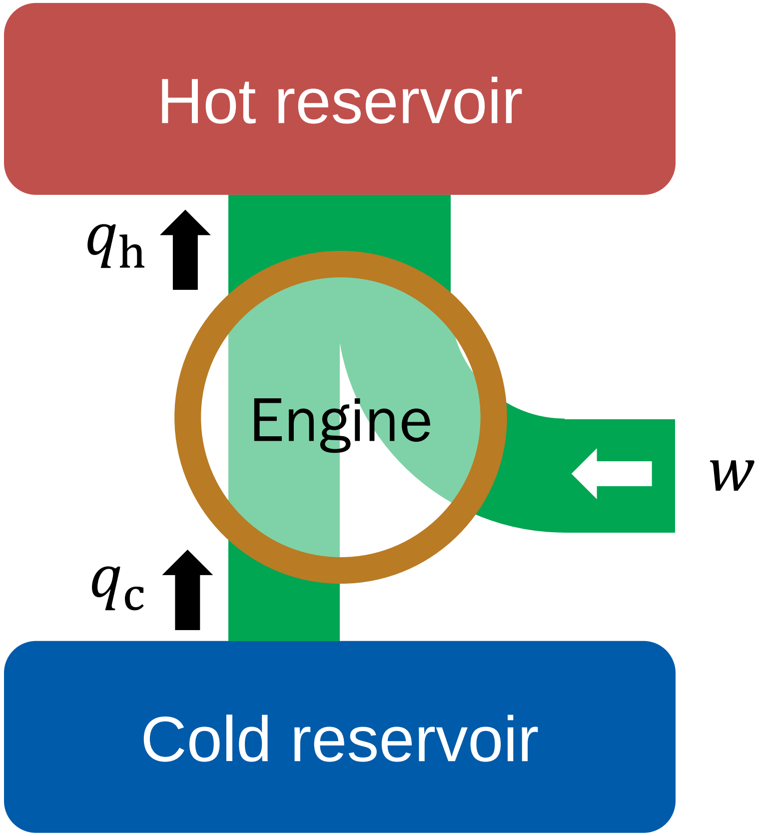
"No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work."

The Kelvin Statement



$$q_\mathrm{h} + w = 0$$

“Refrigerator”

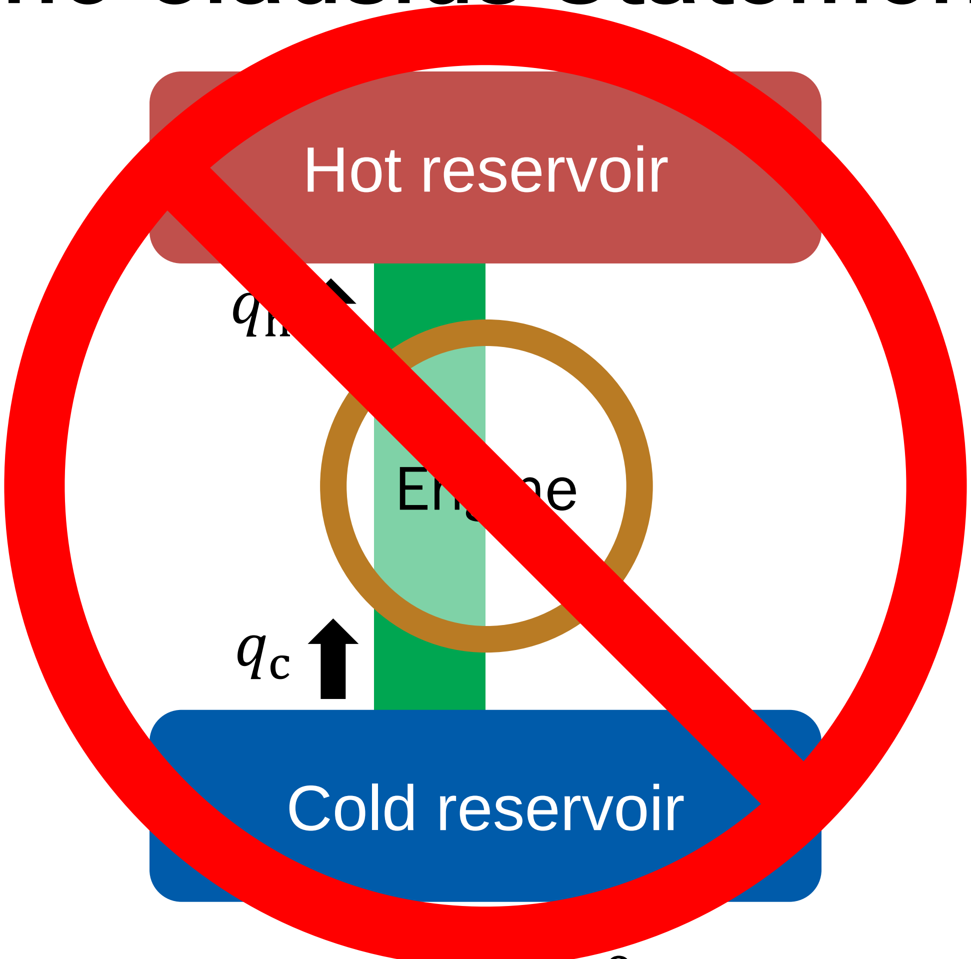


$$q_{\mathrm{h}} + q_{\mathrm{c}} + w = 0$$

Second Law (Clausius)

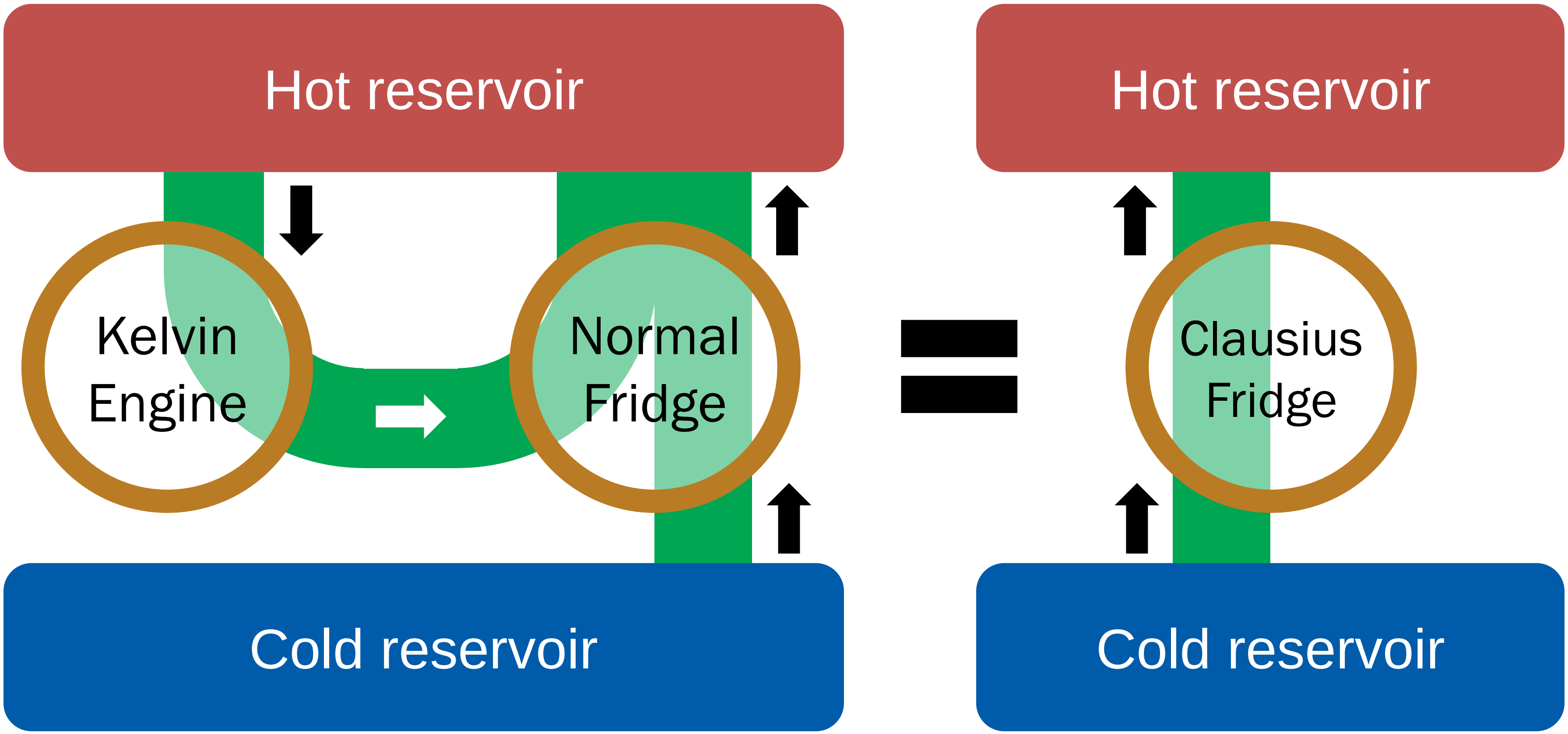
"Heat does not flow spontaneously from a cool body to a hotter body."

The Clausius Statement

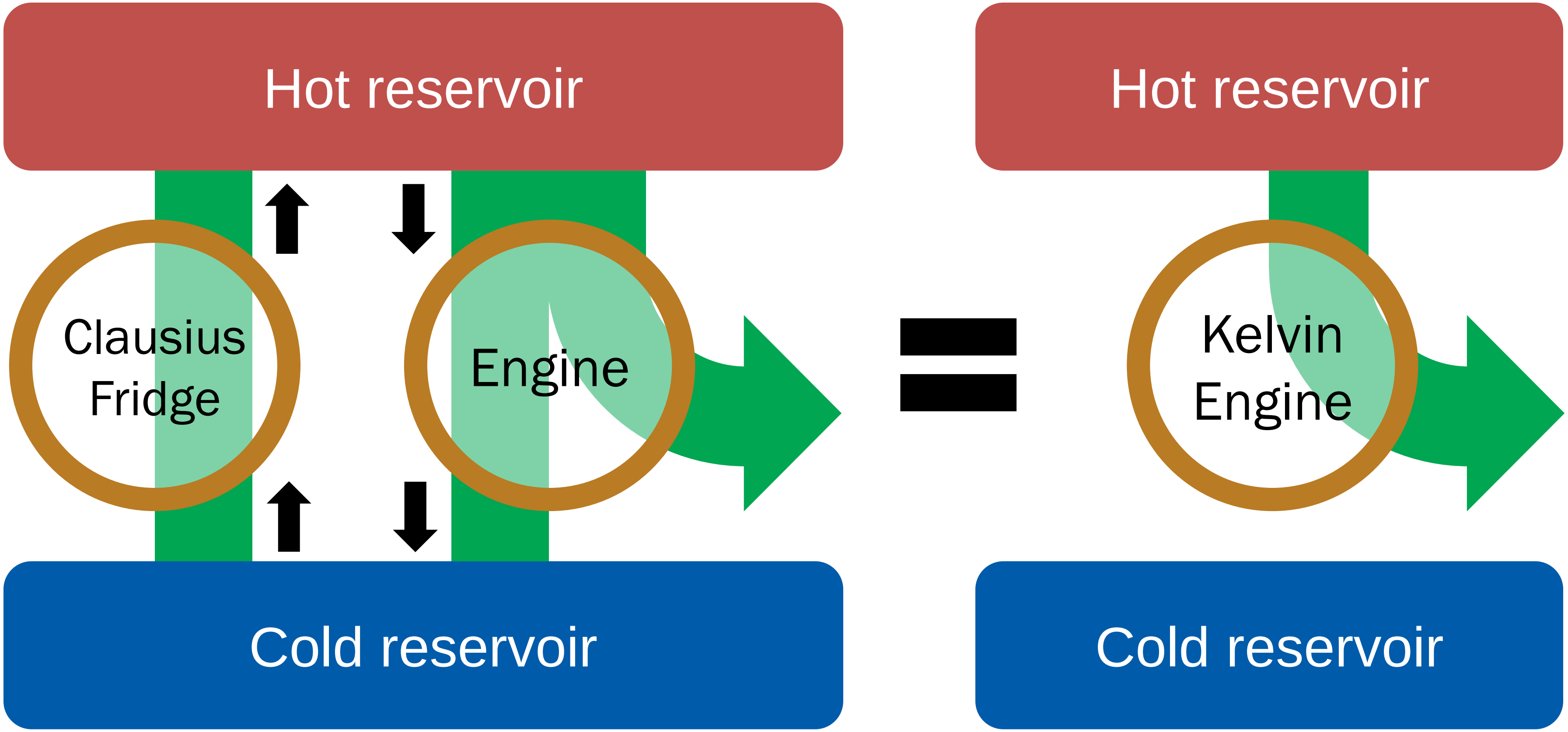


$$q_h + q_c = 0$$

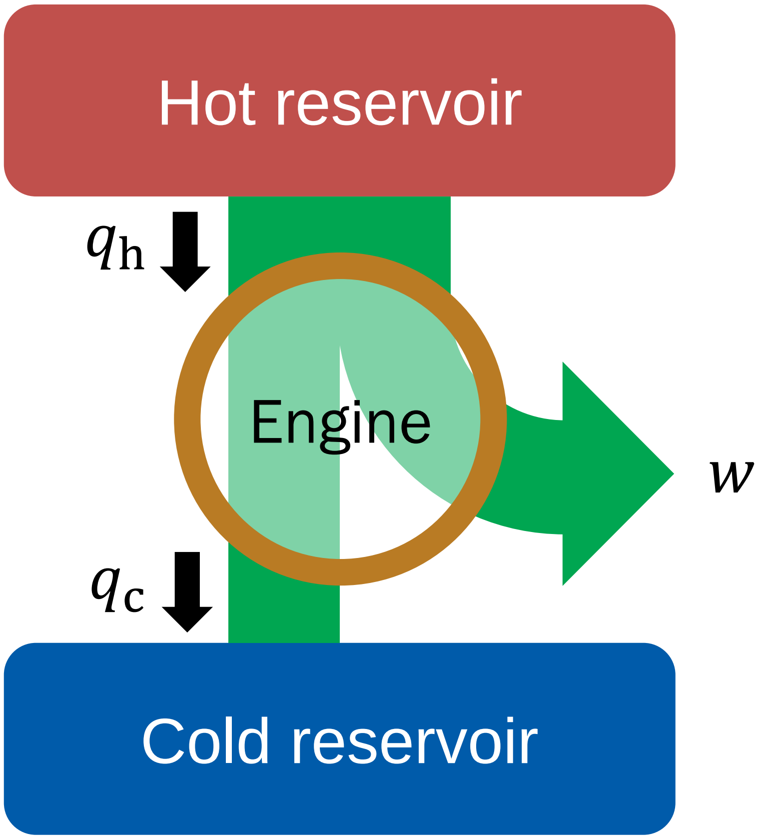
Kelvin → Clausius



Clausius → Kelvin



Efficiency of the Engine



$$q_\mathrm{h} + q_\mathrm{c} + w = 0$$

$$\eta = \frac{|w|}{|q_\mathrm{h}|}$$

Hence,

$$\eta = \frac{|q_\mathrm{h}| - |q_\mathrm{c}|}{|q_\mathrm{h}|} = 1 - \frac{|q_c|}{|q_h|}$$